

ARCHITECT CLOSURE REVIEW

RMA Save Must Support a Product Item Without an Opportunity

Production Case: 00010158

DevDO Tracking Case: 00010704 - Closed

Prepared: September 1, 2026

Environment: Probo Medical DevDO sandbox

Review Status: DEVDO PASS | READY FOR PRODUCTION DEPLOYMENT/UAT

Architect determination: DevDO passed. The implementation is ready for Production deployment and a save-path smoke test using an RMA without an Opportunity. Keep Production case 00010158 open until that test succeeds.

1. Production Case Summary

Field	Verified value
Subject	Need help with saving an RMA in Salesforce - receiving an error
Production status	Claimed
Priority	Medium
Business issue	RMA A-206654 failed during P_RMATrigger processing when an RMA was saved without an assigned Opportunity.

2. Root Cause and Resolution

Root cause: Opportunity Product synchronization dereferenced RepairOpp.Id and Product Item values before confirming that both records existed.

Null-safe guard: Opportunity Product synchronization now runs only when both thisPItem and RepairOpp are non-null.

Valid save path: Selecting a Product Item remains valid even when the RMA has no Opportunity; the unrelated Opportunity synchronization is skipped.

Scope control: No default Opportunity is manufactured, and no relationship is hard-coded or silently inferred.

No-hardcoding control: No Salesforce IDs, Opportunity IDs, Product Item IDs, or fallback business values were added. Runtime relationships determine whether synchronization is applicable.

3. Steps to Reproduce and Validate

Execution boundary: Perform these steps in DevDO with non-production test records. Repeat in Production only after the Change Set is deployed and the business owner authorizes the smoke test.

Preconditions: Use DevDO test data and an authorized RMA user. Have a valid Product Item available, and do not assign an Opportunity to the test RMA.

Step	Action
1	Create a new RMA, or open an editable test RMA, with Opportunity left blank.
2	Select the valid Product Item and complete all other required RMA fields.
3	Enter Quality Issue Notes so the relevant P_RMATrigger processing path is exercised.
4	Save the RMA and refresh the record.
5	Confirm the selected Product Item and notes remain saved while Opportunity remains blank.

Expected result and defect signature: Fixed result: the RMA saves successfully without an Opportunity and without a P_RMATrigger exception. Original defect signature: the save fails while the handler attempts Opportunity Product synchronization.

4. Verification Evidence

Focused regression: P_RMATriggerHandlerTest2.case00010158AllowsProductItemWithoutOpportunity

Acceptance assertion: An RMA saves with its selected Product Item and Quality Issue Notes while Opportunity__c remains null; the transaction completes without the trigger error.

Evidence	Result
Deployment validation	0Afik0000000WpWSAU - 16/16 components deployed; 13/13 deployment-gate tests passed
Final regression rerun	707iK0000005xkHQAQ - 14 tests passed; 0 failures; 100% pass rate
Focused handler coverage	P_RMATriggerHandler focused coverage: 83%
DevDO tracking case	00010704 - Closed and independently read back

5. Story-Specific Change Set Scope

Component type	API name	State
Apex Class	P_RMATriggerHandler	Modified
Apex Test	P_RMATriggerHandlerTest2	Regression assertion

Outbound Change Set: DEVDO Next 4 High Priority Fixes (0A2iK00000001lh) - 16 components; not uploaded

Production safety: The Change Set has not been uploaded or deployed. Production metadata and Production case status were not changed.

6. Closure Gate

Gate	Status	Required evidence
DevDO implementation complete	YES	Components deployed and active in DevDO
Automated regression passed	YES	Test job 707iK0000005xkHQAQ
Outbound Change Set prepared	YES	0A2iK00000001lh; 16 components
Production deployment	NO - PENDING	Upload, validate, and deploy the Change Set
Production business smoke test	NO - PENDING	Execute the case-specific acceptance path
Close original Production case	HOLD	Close only after both Production gates pass

Final recommendation: DevDO work is complete and the sandbox tracking case is correctly closed. Move the original Production case to Ready for Production/UAT; do not mark the Production issue resolved until deployment and smoke-test evidence are attached.